

RESEARCH PUBLICATIONS

Pietro Coretto

Wednesday 27th May, 2026

Papers in journals (with peer-review)

- Coretto, P. and C. Hennig (2025). Consistency for constrained maximum likelihood estimation and clustering based on mixtures of elliptically-symmetric distributions under general data generating processes. *Journal of Multivariate Analysis*, vol. 209: page 105446. URL: <https://doi.org/10.1016/j.jmva.2025.105446>. Open access.
- Coraggio, L. and P. Coretto (2024). Asymptotic results for the estimation of the quadratic score of a clustering. *Mathematics*, vol. 12(21).
- Coretto, P. (2024). Discussion (invited) of the paper “connecting model-based and model-free approaches to linear least squares regression” by Lutz Dümbgen and Laurie Davies (2024). *Statistica*, vol. 84(2): page 105.
- Coraggio, L. and P. Coretto (2023). Selecting the number of clusters, clustering models, and algorithms. A unifying approach based on the quadratic discriminant score. *Journal of Multivariate Analysis*, vol. 196: pages Paper No. 105181, 20. URL: <https://doi.org/10.1016/j.jmva.2023.105181>.
- Hennig, C. and P. Coretto (2022). An adequacy approach for deciding the number of clusters for OTRIMLE robust Gaussian mixture-based clustering. *Australian & New Zealand Journal of Statistics*, vol. 64(2): page 230.
- Coretto, P. (2022). Estimation and computations for Gaussian mixtures with uniform noise under separation constraints. *Statistical Methods & Applications. Journal of the Italian Statistical Society*, vol. 31(2): page 427.
- Giordano, F. and P. Coretto (2020). A monte carlo subsampling method for estimating the distribution of signal-to-noise ratio statistics in nonparametric time series regression models. *Statistical Methods & Applications*, vol. 29(3): page 483.
- Coretto, P. (2020). Discussion (invited) of: Multiple-systems analysis for the quantification of modern slavery: classical and bayesian approaches; by Bernard W. Silverman. *Journal of the Royal Statistical Society, Series A (Statistics in Society)*, vol. 183(3): page 725.
- Coretto, P., M. La Rocca, and G. Storti (2020). Improving many volatility forecasts using cross-sectional volatility clusters. *Journal of Risk and Financial Management*, vol. 13(4): page 1.
- Storti, G. and P. Coretto (2020). Discussion (invited) of: Linear mixed effects models for non-gaussian continuous repeated measurement data; by O. Asar, D. Bolin, P. J. Diggle and J. wallin. *Journal of the Royal Statistical Society, Series C (Applied Statistics)*, vol. 69(5): page 46. URL: <https://doi.org/10.1111/rssc.12405>.
- Serra, A., S. Önlü, P. Coretto, and D. Greco (2019). An integrated quantitative structure and mechanism of action-activity relationship model of human serum albumin binding. *Journal of Cheminformatics*, vol. 11(38): page 1.
- Serra, A., P. Coretto, M. Fratello, and R. Tagliaferri (2018). Robust and sparse correlation matrix estimation for the analysis of high-dimensional genomics data. *Bioinformatics*, vol. 34(4): page 625.

- Coretto, P., A. Serra, and R. Tagliaferri (2018). Robust clustering of noisy high-dimensional gene expression data for patients subtyping. *Bioinformatics*, vol. 34(23): page 4064.
- Coretto, P. and C. Hennig (2017). Consistency, breakdown robustness, and algorithms for robust improper maximum likelihood clustering. *Journal of Machine Learning Research*, vol. 18(142): page 1.
- Coretto, P. and F. Giordano (2017). Nonparametric estimation of the dynamic range of music signals. *Australian & New Zealand Journal of Statistics*, vol. 59(4): page 389.
- Coretto, P. and C. Hennig (2016). Robust improper maximum likelihood: tuning, computation, and a comparison with other methods for robust gaussian clustering. *Journal of the American Statistical Association*, vol. 111(516): page 1648.
- Coretto, P. (2013). Discussion (invited) of: How to find an appropriate clustering for mixed-type variables with application to socio-economic stratification; by C. Hennig and T. F. Liao. *Journal of the Royal Statistical Society, Series C (Applied Statistics)*, vol. 62: page 346.
- Coretto, P. and C. Hennig (2011). Maximum likelihood estimation of heterogeneous mixtures of gaussian and uniform distributions. *Journal of Statistical Planning and Inference*, vol. 141(1): page 462.
- Coretto, P. and C. Hennig (2010). A simulation study to compare robust clustering methods based on mixtures. *Advances in Data Analysis and Classification*, vol. 4(2): page 111.

Papers in book collections (with peer-review)

Papers in conference proceedings (with peer-review)

- Coraggio, L. and P. Coretto (2023a). Quadratic discriminant scoring for selecting clustering solutions. In *Statistical methods for evaluation and quality: techniques, technologies and trends (T3). Book of Short Papers IES 2023*, page 355.
- Coraggio, L. and P. Coretto (2023b). Empirical analysis of the quadratic scoring for selecting clustering solutions. In *Book of Abstracts and Short Papers, 14th Scientific Meeting of the Classification and Data Analysis Group*, page 398.
- Coretto, P. and C. Hennig (2021). Non-parametric consistency for the Gaussian mixture maximum likelihood estimator. In *Cladag 2021 - Book of short Papers*, page 116.
- Coretto, P. and L. Coraggio (2021). In-sample and cross-validated likelihood-type criteria for clustering selection. In *Book of Short Papers of the 5th international workshop on Models and Learning for Clustering and Classification*, page 27.
- Coretto, P., A. Serra, and R. Tagliaferri (2019). Noise resistant clustering of high-dimensional gene expression data. In *Cladag 2019 - Book of Short Papers*, page 132.
- Coraggio, L. and P. Coretto (2019). Likelihood-type methods for comparing clustering solutions. In *Cladag 2019 - Book of Short Papers*, page 120.
- Serra, A., P. Coretto, and R. Tagliaferri (2017). On the noisy high-dimensional gene expression data analysis. In A. Petrucci and R. Verde, editors, *Statistics and data science: new challenges, new generations*, page 919.
- Coretto, P. and C. Hennig (2015). Robust model-based clustering with covariance matrix constraints. In *10th meeting of Classification and Data Analysis Group*, page 1.
- Coretto, P. and C. Hennig (2014). Robust covariance matrix estimation with regularization. In S. I. di Statistica, editor, *Proceedings of the 47th Scientific Meeting of the Italian Statistical Society*, page 1.

- Coretto, P. (2012). Adaptive robust location–scale estimation. In S. I. di Statistica, editor, *Proceedings of the XLVI Scientific Meeting*, page 1.
- Coretto, P., M. La Rocca, and G. Storti (2009). Group structured volatility. In *Book of Short Papers CLADAG09 Catania*, page 119.
- Coretto, P. and C. Hennig (2009). Robust model-based clustering using an improper constant density selected by a distance optimizatin for non-outliers. In *JSM Proceedings, IMS General Theory Section*, page 3847.
- Coretto, P. (2008). Robust mixture modelling. In *Joint Meeting of the SFC and CLADAG*, page 69.
- Coretto, P. and C. Hennig (2007). Choice of the improper density in robust improper ML for finite normal mixtures. In *Classification and Data Analysis 2007*, page 439.